

RITWIK GUPTA

Principal Systems & Software Engineer | ritwikg.205@gmail.com | San Francisco, CA

GitHub: github.com/ritwikg | LinkedIn: linkedin.com/in/ritwikg

SUMMARY

Systems engineer with 8+ years of experience in low-latency C++/Rust, memory lock-free algorithms, distributed consensus protocols (Raft/Paxos), and high-throughput GPU compute pipelines.

CORE COMPETENCIES & SKILLS

Systems Languages: C++20/23, Rust (Tokio/Async), Go, C, Assembly (SIMD/AVX-512), CUDA

Distributed Infrastructure: Raft & Paxos Consensus, gRPC, Protocol Buffers, Kafka, LSM-Trees

Cloud & OS: Linux Kernel Tuning, eBPF, Docker, Kubernetes, Lock-Free Memory Allocators

Web & Runtime: TypeScript, Node.js, WebAssembly (Wasm), React, WebSockets, Three.js

PROFESSIONAL EXPERIENCE

Principal Systems Architect - High-Frequency Systems Engine

2023 - Present | San Francisco, CA

- Reduced end-to-end P99 transaction latency from 450 microseconds to 68 microseconds using zero-copy ring buffers and lock-free concurrent queues.
- Built parallel CUDA risk analysis kernels processing over 2.4 million evaluations per second.

Senior Distributed Systems Engineer - Aether Cloud

2020 - 2023 | San Francisco, CA

- Implemented sharded Raft consensus protocol handling 15,000 requests/sec with log compaction.
- Designed custom arena memory allocators, cutting system memory fragmentation by 35%.

EDUCATION

Bachelor of Science in Computer Science

Focus on Operating Systems & Parallel Computing | Honors Graduate